Innovative Non-Invasive Treatment for Perinatal Depression : A Korean Perspective

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Introduction

Perinatal depression significantly affects maternal and child health, necessitating effective and safe interventions. Conventional treatments raise concerns about potential side effects, especially during pregnancy and breastfeeding. This study investigates **Transcranial Direct Current Stimulation (tDCS)** as a non-invasive, safe alternative for perinatal depression treatment, leveraging its ability to modulate cortical excitability.

Objectives

To evaluate the **effectiveness of tDCS** in alleviating **depression among perinatal women**, providing insights into its potential as a standard treatment in Korean healthcare settings.

Materials and Methods

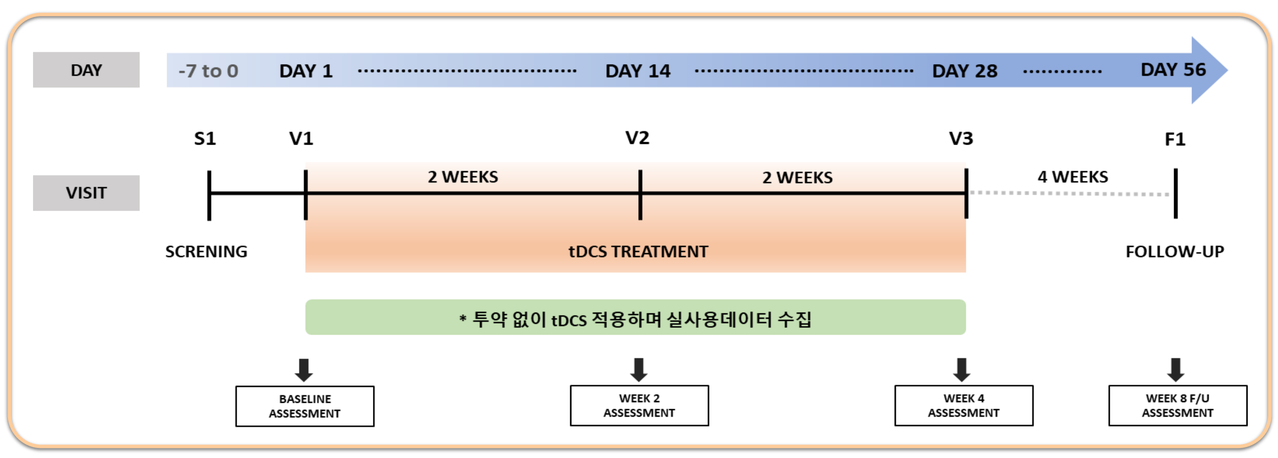
- **Participants: 22** perinatal women from 4 hospitals in Korea
- **Intervention:** Daily **30-minute sessions** of **2mA tDCS** for **4** weeks
- Depression levels measured pre- and post-treatment using:
  - **CESD-R** : Center for Epidemiologic Studies Depression Scale Revised
  - **BDI-II** : Beck Depression Inventory, Second Edition
  - **MADRS** : Montgomery-Åsberg Depression Rating Scale
  - **K-EPDS** : Korean version of the Edinburgh Postnatal Depression Scale
  - **DASS-21 :** Depression Anxiety Stress Scales-21
- **Analysis Tools:** SPSS v29.0.1.1 with paired t-tests and descriptive statistics

◆ **Non-invasive brain stimulation**

Current strength was set to 2 mA and the duration of each stimulation was 30 min

◆ **Treatment Procedure**

1. Place the band on the appropriate area of the forehead and turn on the module.
2. Ensure the patch is firmly attached to the forehead without lifting from the skin.



Results

Table 1. Demographic characteristics of the participants (n=22)

	Mean	SD
Age (years)	35.05	4.89
Height (cm)	163.81	6.89
Prepregnant weight (kg)	61.44	10.72
Current weight (kg)	66.54	14.21
Prepregnant BMI (kg/m^2)	22.77	4.25
Current BMI (kg/m^2)	24.80	5.14

Table 2. Pre- and post-intervention comparisons of psychological test scores (n=22)

		Mean	SD	t	p
CESD-R	Pre	8.82	5.54	3.263	**0.002**
	Post	5.27	4.50		
BDI-II	Pre	25.23	10.19	4.566	**<0.001**
	Post	18.68	11.32		
MADRS	Pre	21.55	5.41	3.681	**<0.001**
	Post	14.64	9.76		
EPDS	Pre	16.0	6.66	4.536	**<0.001**
	Post	10.82	6.96		

Table 3. Comparison of sub-scales of DASS-21 before and after tDCS treatment in perinatal women (n=22)

		Mean	SD	t	p
Depression	Pre	7.62	5.01	4.352	**<0.001**
	Post	4.66	4.59		
Anxiety	Pre	6.59	4.72	5.406	**<0.001**
	Post	3.79	3.76		
Stress	Pre	9.10	4.50	4.115	**<0.001**
	Post	6.38	5.00		

Conclusion

This study reinforces **tDCS** as a **promising, safe, and effective intervention** for reducing perinatal depression levels. Findings emphasize the importance of culturally contextualized, non-invasive treatments for maternal mental health.

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